

# Cryptography and Network Security Overview & Chapter 1

Fifth Edition  
by William Stallings

# Background

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- Information Security requirements have changed in recent times
- traditionally provided by physical and administrative mechanisms
- computer use requires automated tools to protect files and other stored information
- use of networks and communications links requires measures to protect data during transmission

# Definitions

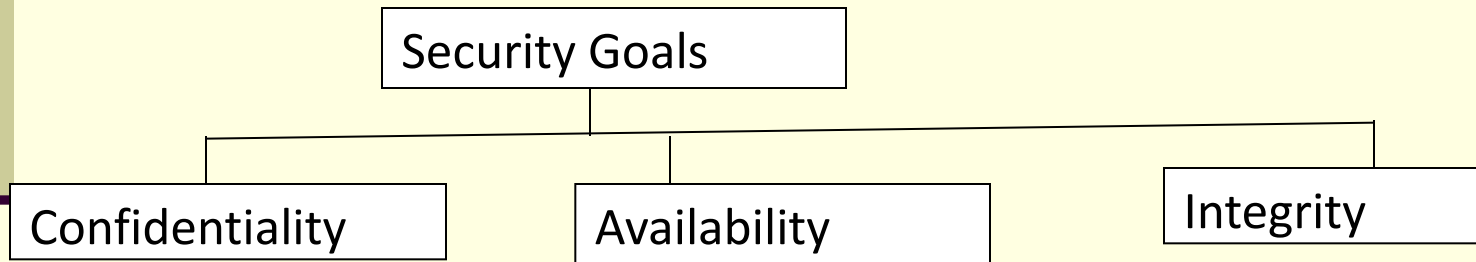
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- **Computer Security** - generic name for the collection of tools designed to protect data and to thwart hackers.
- **Network Security** - measures to protect data during their transmission.
- **Internet Security** - measures to protect data during their transmission over a collection of interconnected networks.

# Security Goals

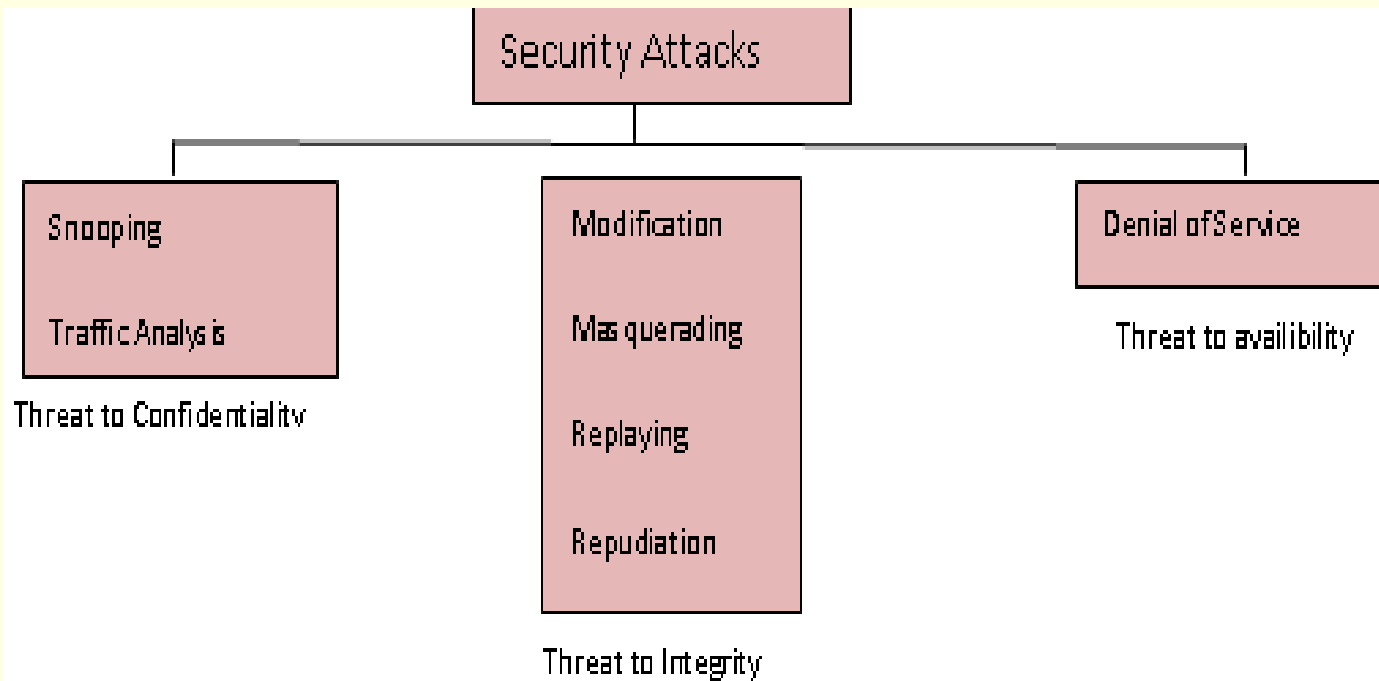
Information needs to be hidden from

- Unauthorized access (Confidentiality)
- Protected from unauthorized change (integrity)
- Available to an authorized entity when it is needed (availability)



# Security Attacks

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# Attacks Threatening Confidentiality

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- Snooping

- Unauthorized access to or interception of data.

- Traffic analysis

- Can find some other type information by monitoring online traffic.

- He can find the electronic address (such as e-mail address) of the sender or the receiver.

# Attacks Threatening Integrity

- **Modification**

- After intercepting or accessing information, the attacker modifies the information to make it beneficial to himself

- **Masquerading**

- Masquerading or spoofing(trick) happens when the attacker impersonates somebody else.

- **Replaying**

- The attacker obtains a copy of a message sent by a user and later tries to replay it .

- **Repudiation**

- Performed by one of the two parties in the communication: the sender or the receiver.
  - - The sender of the message might later deny that she has sent the message.
  - - The receiver of the message might later deny that he has received the message

# Attacks Threatening Availability

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- It may slow down or totally interrupt the service of a system.
- He might send so many bogus requests to a server that the server crashes because of the heavy load.
- The attacker might intercept and delete a server's response to a client, making the client to believe that the server is not responding.

# Levels of Impact

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- can define 3 levels of impact from a security breach
  - Low
  - Moderate
  - High

# Low Impact

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- The loss could be expected to have a limited adverse effect on organizational operations, organizational assets, or individuals.
- A limited adverse effect means that, for example, the loss of confidentiality, integrity, or availability might
  - (i) cause a degradation in mission capability to an extent and duration that the organization is able to perform its primary functions, but the effectiveness of the functions is noticeably reduced;
  - (ii) result in minor damage to organizational assets;
  - (iii) result in minor financial loss; or
  - (iv) result in minor harm to individuals.

# Moderate Impact

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- The loss could be expected to have a serious adverse effect on organizational operations, organizational assets, or individuals.
- A serious adverse effect means that, for example, the loss might
  - (i) cause a significant degradation in mission capability to an extent and duration that the organization is able to perform its primary functions, but the effectiveness of the functions is significantly reduced;
  - (ii) result in significant damage to organizational assets;
  - (iii) result in significant financial loss; or
  - (iv) result in significant harm to individuals that does not involve loss of life or serious, life-threatening injuries.

# High Impact

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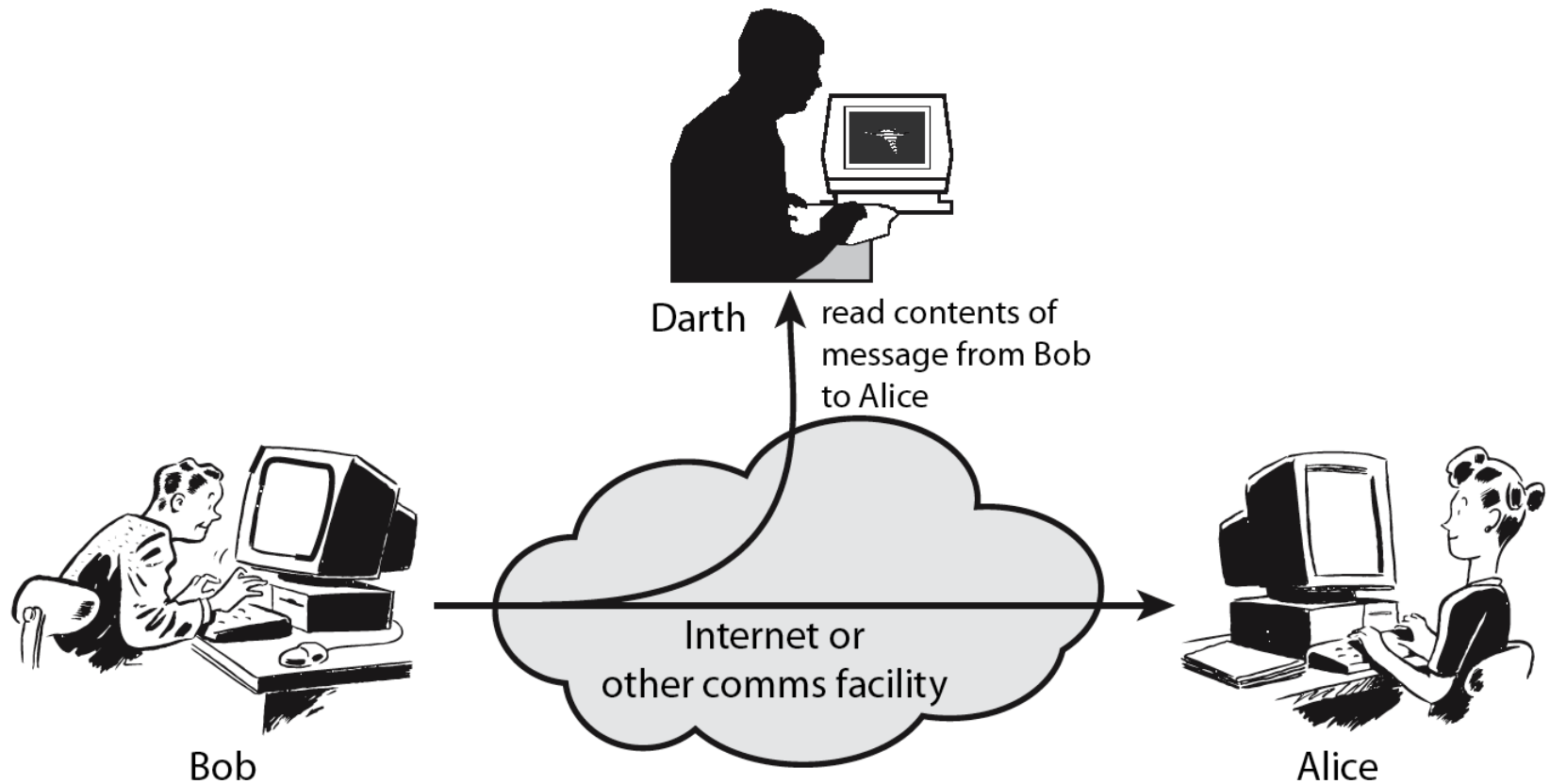
- The loss could be expected to have a severe or catastrophic adverse effect on organizational operations, organizational assets, or individuals.
- A severe or catastrophic adverse effect means that, for example, the loss might
  - (i) cause a severe degradation in or loss of mission capability to an extent and duration that the organization is not able to perform one or more of its primary functions;
  - (ii) result in major damage to organizational assets;
  - (iii) result in major financial loss; or
  - (iv) result in severe or catastrophic harm to individuals involving loss of life or serious life threatening injuries.

# Security Attack

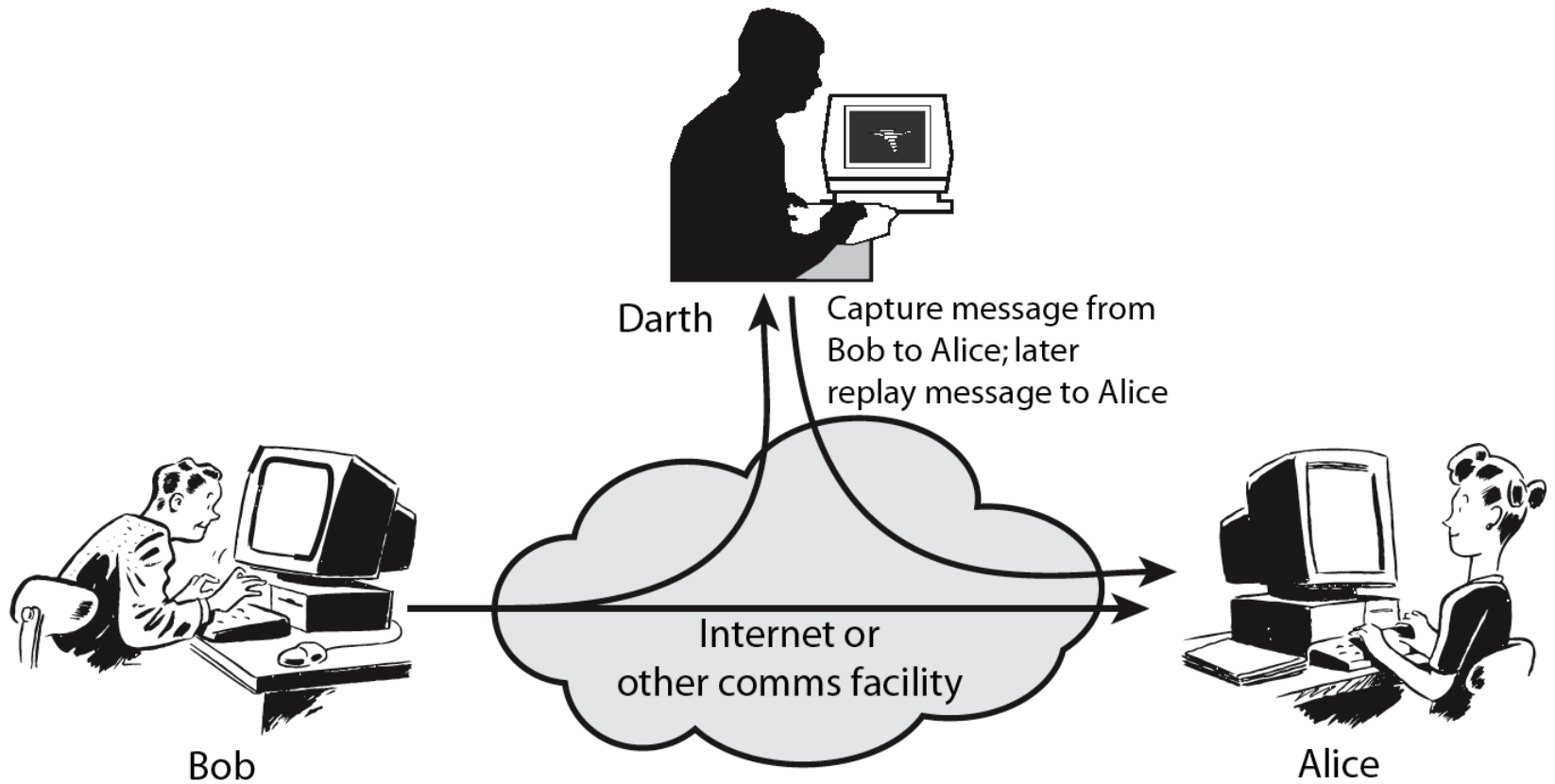
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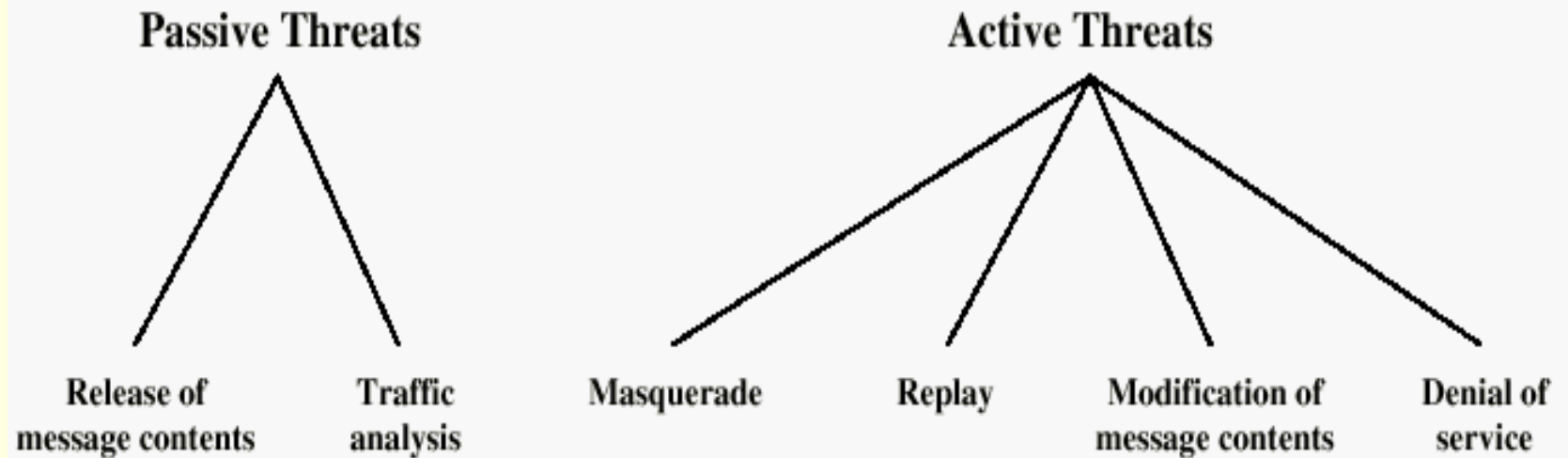
- any action that compromises the security of information owned by an organization
- information security is about how to prevent attacks, or failing that, to detect attacks on information-based systems
- often *threat* & *attack* used to mean same thing
- have a wide range of attacks
- can focus on generic types of attacks
  - passive
  - active

# Passive Attacks



# Active Attacks





**Figure 1.2 Active and Passive Security Threats**

# Security Service

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- enhance security of data processing systems and information transfers of an organization
- intended to counter security attacks
- using one or more security mechanisms
- often replicates functions normally associated with physical documents
  - which, for example, have signatures, dates; need protection from disclosure, tampering, or destruction; be notarized or witnessed; be recorded or licensed

# Security Services

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- X.800:

“a service provided by a protocol layer of communicating open systems, which ensures adequate security of the systems or of data transfers”

- RFC 2828:

“a processing or communication service provided by a system to give a specific kind of protection to system resources”

# Security Services (X.800)

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- **Authentication** - assurance that the communicating entity is the one claimed
- **Access Control** - prevention of the unauthorized use of a resource
- **Data Confidentiality** –protection of data from unauthorized disclosure
- **Data Integrity** - assurance that data received is as sent by an authorized entity
- **Non-Repudiation** - protection against denial by one of the parties in a communication

# Security Mechanism

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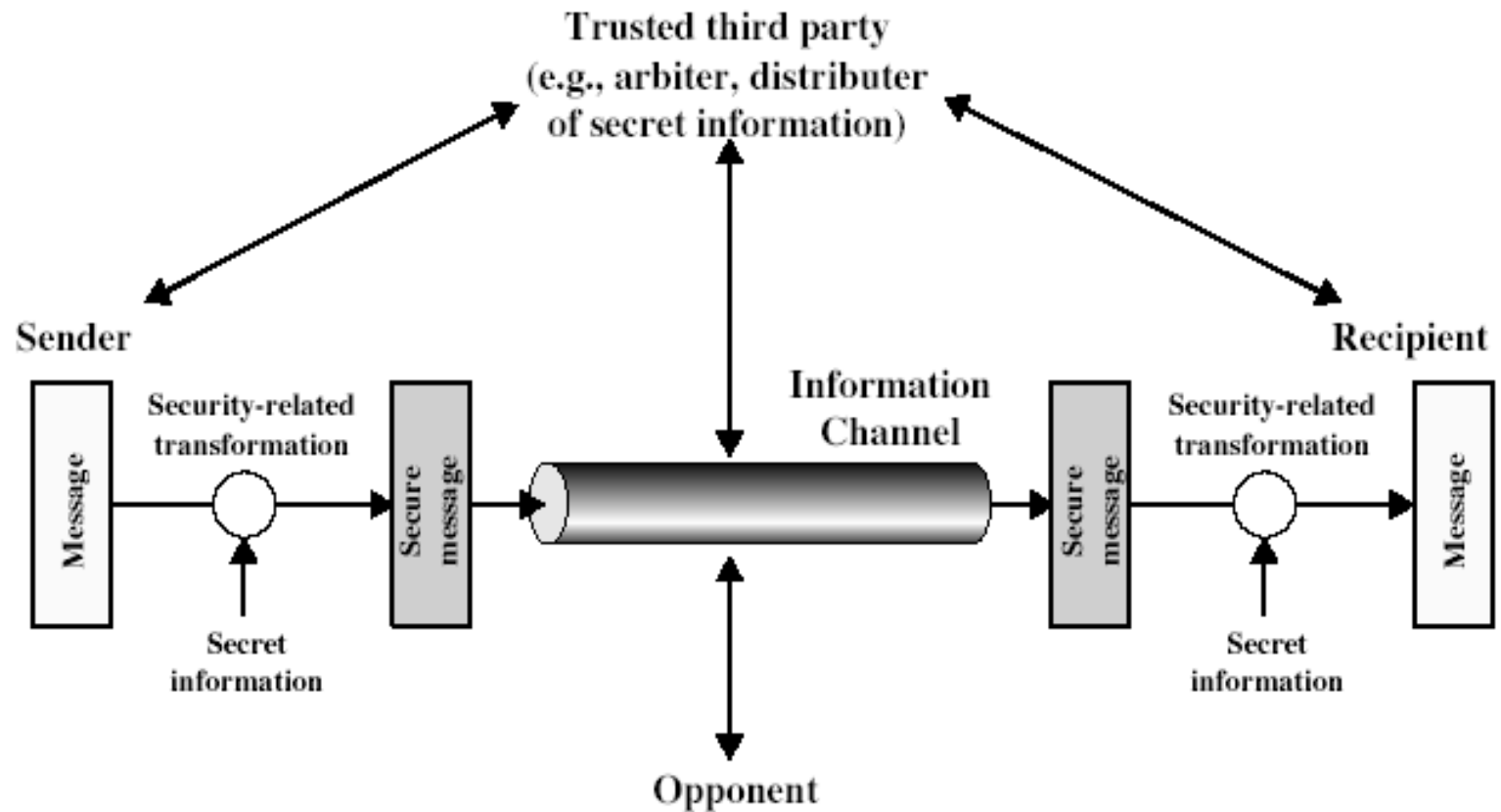
- feature designed to detect, prevent, or recover from a security attack
- no single mechanism that will support all services required
- however one particular element underlies many of the security mechanisms in use:
  - **cryptographic techniques**
- hence our focus on this topic

# Security Mechanisms (X.800)

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- specific security mechanisms:
  - encipherment, digital signatures, access controls, data integrity, authentication exchange, traffic padding, routing control, notarization
- pervasive security mechanisms:
  - trusted functionality, security labels, event detection, security audit trails, security recovery

# Model for Network Security

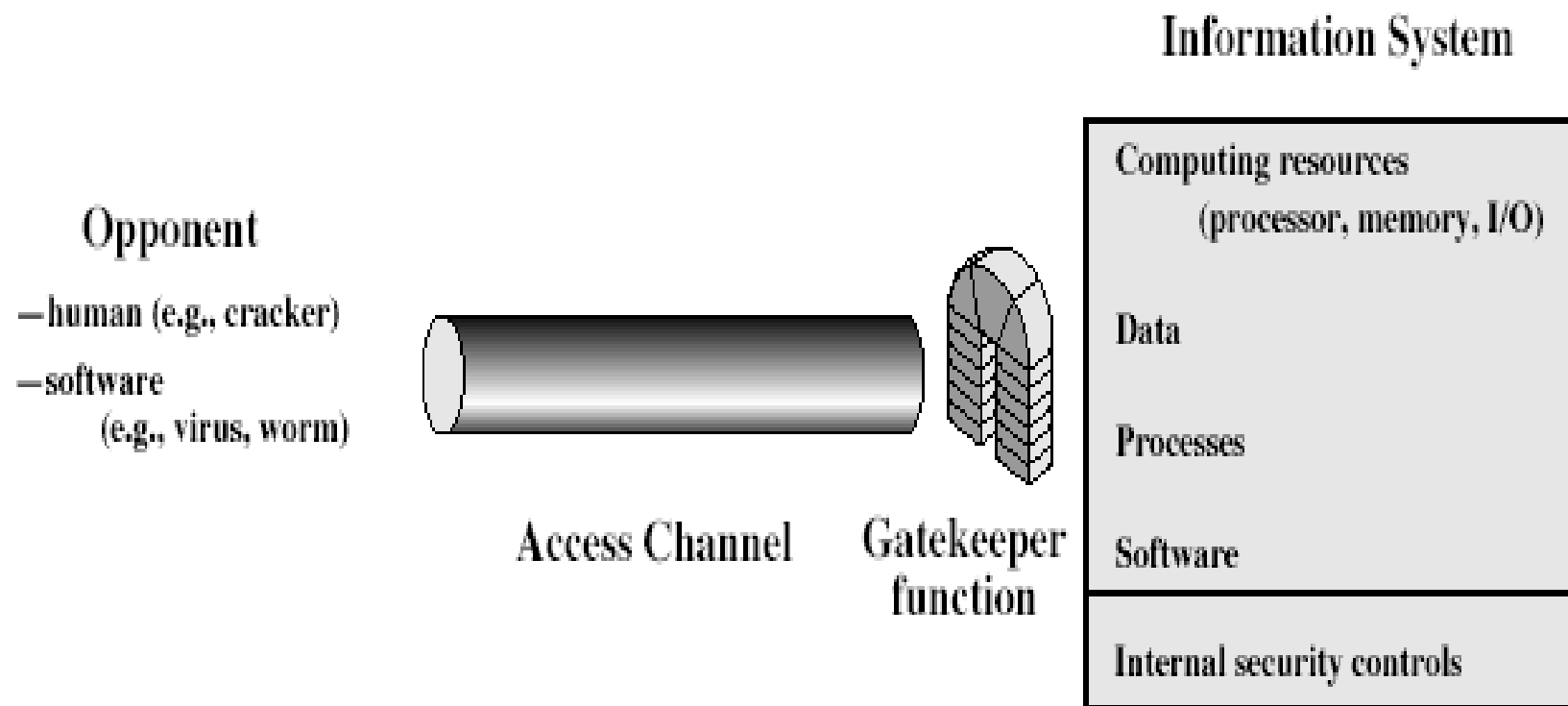


# Model for Network Security

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- using this model requires us to:
  1. design a suitable algorithm for the security transformation
  2. generate the secret information (keys) used by the algorithm
  3. develop methods to distribute and share the secret information
  4. specify a protocol enabling the principals to use the transformation and secret information for a security service

# Model for Network Access Security



# Model for Network Access Security

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- using this model requires us to:
  1. select appropriate gatekeeper functions to identify users
  2. implement security controls to ensure only authorised users access designated information or resources
- trusted computer systems may be useful to help implement this model



# Chapter 2

## Classical Encryption Techniques

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# Cryptography

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- Cryptography is the study of
  - **Secret** (crypto-) **writing** (-graphy)
- Concerned with developing algorithms:
  - Conceal the context of some message from all except the sender and recipient (**privacy** or **secrecy**), and/or
  - Verify the correctness of a message to the recipient (**authentication**)
  - Form the basis of many technological solutions to computer and communications security problems

# Cryptography

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- Basic idea: convert **clear text** (also called **plain text** – the original message) to **ciphertext** (the encrypted message)

$\text{ciphertext} = \text{encrypt}(\text{plaintext}, K_E)$

$\text{plaintext} = \text{decrypt}(\text{ciphertext}, K_C)$

- Can either make the encryption process hidden, so that an intruder cannot know it
- Or, can use a known technique and use a hidden **key**

# Some Basic Terminology

- **plaintext** - original message
- **ciphertext** - coded message
- **cipher** - algorithm for transforming plaintext to ciphertext
- **key** - info used in cipher known only to sender/receiver
- **encipher (encrypt)** - converting plaintext to ciphertext
- **decipher (decrypt)** - recovering plaintext from ciphertext
- **cryptography** - study of encryption principles/methods
- **cryptanalysis (codebreaking)** - study of principles/ methods of deciphering ciphertext *without* knowing key
- **cryptology** - field of both cryptography and cryptanalysis

# Cryptography

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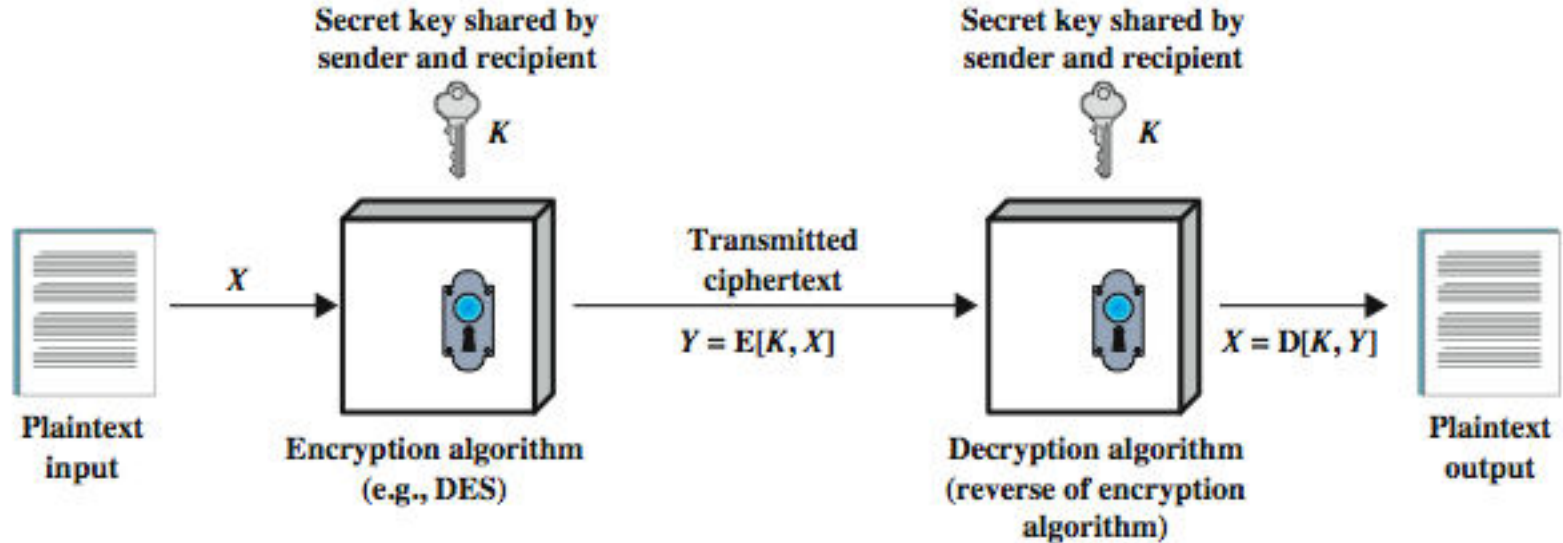
- characterize cryptographic system by:
  - type of encryption operations used
    - substitution / transposition / product
  - number of keys used
    - single-key or private / two-key or public
  - way in which plaintext is processed
    - block / stream

# Number of keys used (Secret-Key Cryptography)

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- traditional **secret/single key** cryptography uses **one** key
- shared by both sender and receiver
- if this key is disclosed communications are compromised
- also is **symmetric**, parties are equal
- hence does not protect sender from receiver forging a message & claiming is sent by sender

# Symmetric Encryption



# Public-Key Cryptography

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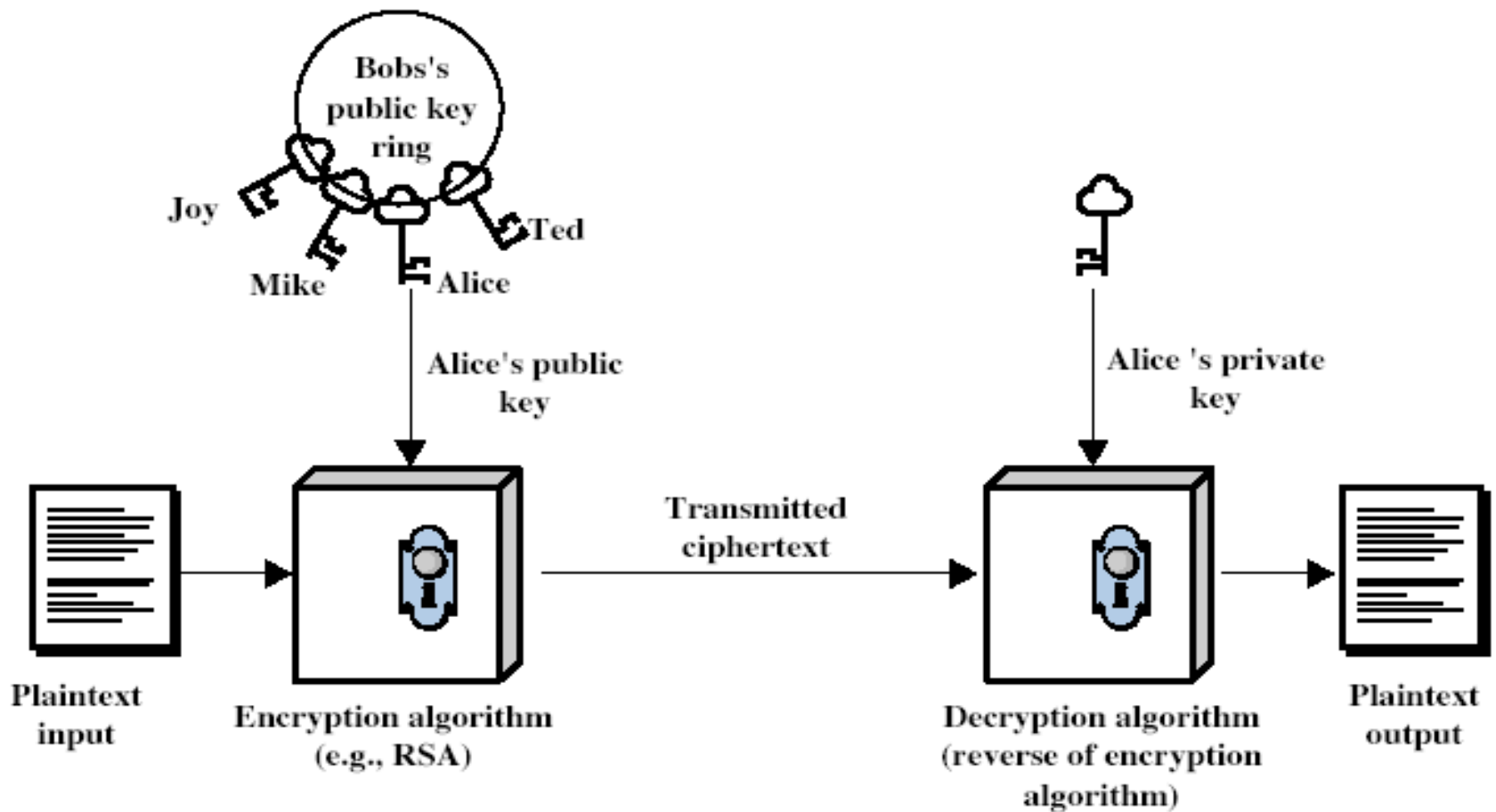
- probably most significant advance in the 3000 year history of cryptography
- uses **two** keys – a public & a private key
- **asymmetric** since parties are **not** equal
- uses clever application of number theoretic concepts to function
- complements **rather than** replaces secret key cryptography

# Public-Key Cryptography

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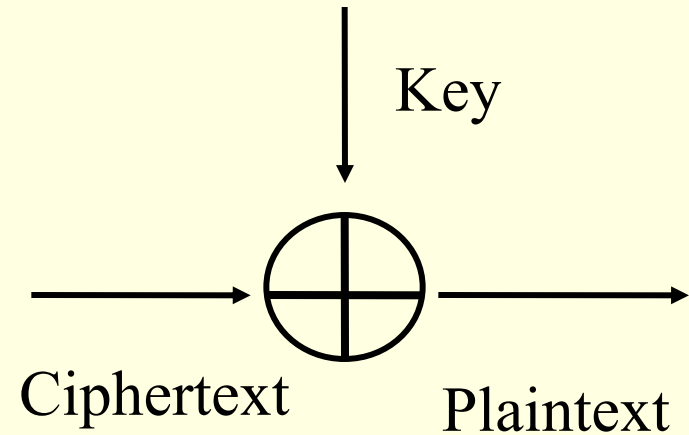
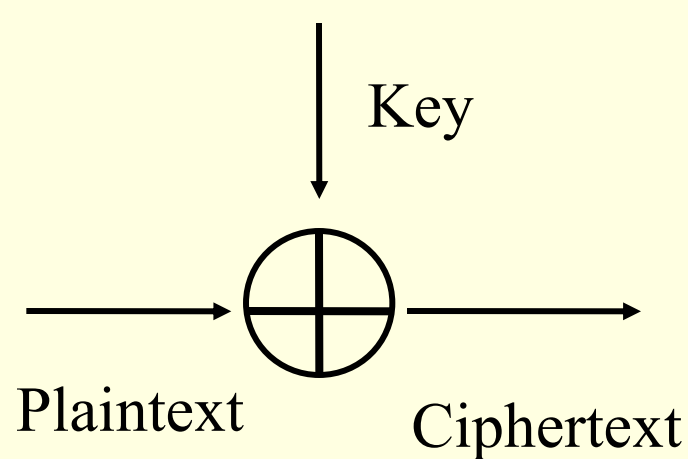
- **public-key/two-key/asymmetric** cryptography involves the use of **two** keys:
  - a **public-key**, which may be known by anybody, and can be used to **encrypt messages**, and **verify signatures**
  - a **private-key**, known only to the recipient, used to **decrypt messages**, and **sign** (create) **signatures**
- is **asymmetric** because
  - those who encrypt messages or verify signatures **cannot** decrypt messages or create signatures

# Public-Key Cryptography



# Way in which plaintext is processed

- Simplest Stream Cipher



# Simple Secret-Key Example

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- P = “abra” which has the binary representation:  
01100001011000100011100101100001

Choose a random string of bits as the key

- 10011101010010001111010101011100

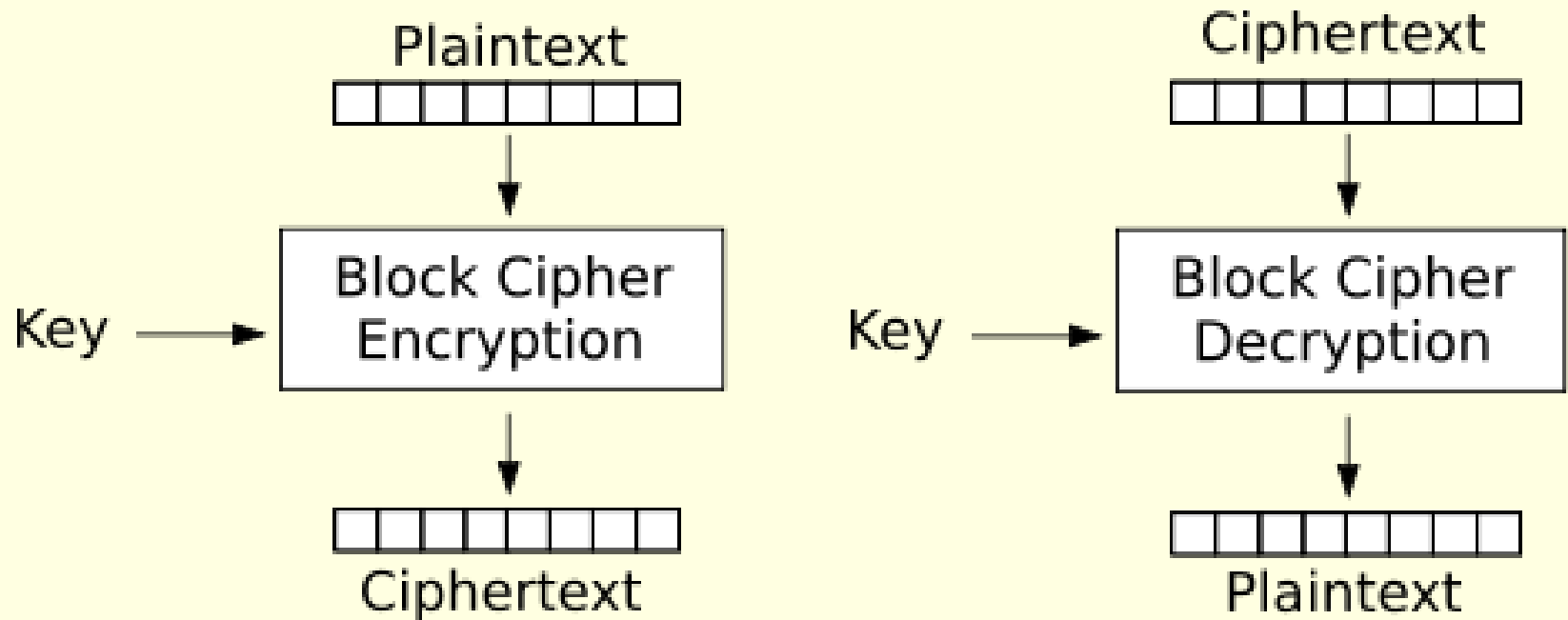
Can use a simple XOR of the binary to get C

- 11111100001010101000011100111101

To get P back, use the same algorithm and key!

The most popular secret key encryption today is DES.

# •Block Cipher



# Type of encryption operations used

## Substitution Technique

### Caesar Cipher

- replaces each letter by 3rd letter on
- can define transformation as:

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w |
| x | y | z |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| A | B | C |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

- mathematically give each letter a number

|    |    |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| a  | b  | c | d | e | f | g | h | i | j | k  | l  | m  | n  | o  | p  | q  | r  | s  | t  | u  | v  | w  | x  |
| y  | z  |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 0  | 1  | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
| 24 | 25 |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

- then have Caesar cipher as:

$$c = E(p) = (p + k) \bmod (26)$$

$$p = D(c) = (c - k) \bmod (26)$$

# Playfair Cipher

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|   |     |   |   |   |
|---|-----|---|---|---|
| s | i/j | m | p | l |
| e | a   | b | c | d |
| f | g   | h | k | n |
| o | q   | r | t | u |
| v | w   | x | y | z |

Key: simple

# Playfair Cipher

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- Use filler letter to separate repeated letters
- Encrypt two letters together
  - Same row— followed letters
    - ac--bd
  - Same column— letters under
    - qw--wi
  - Otherwise—square's corner at same row
    - ar--bq

# Hill Cipher

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- takes  $m$  successive plaintext letters and substitutes for them  $m$  ciphertext letters
- substitution determined by  $m$  liner equations,  $C=KP \text{ mod } 26$
- This example will rely on some linear algebra and some number theory. The 'key' for a hill cipher is a matrix e.g.

$$\begin{bmatrix} 2 & 4 & 5 \\ 9 & 2 & 1 \\ 3 & 8 & 7 \end{bmatrix}$$

- In the above case, we have taken the size to be 3x3, however it can be any size (as long as it is square).
- We now take 3 characters from our plaintext e.g. 'DEF' and create a vector that corresponds to the letters (replace 'A' with 0, 'B' with 1 etc.) to get: [3 4 5] (this is ['D' 'E' 'F']).
- To get our ciphertext we perform a matrix multiplication

$$\begin{bmatrix} 2 & 4 & 5 \\ 9 & 2 & 1 \\ 3 & 8 & 7 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 47 \\ 40 \\ 76 \end{bmatrix} \pmod{26} = \begin{bmatrix} 21 \\ 14 \\ 24 \end{bmatrix} = \text{'VOY'}$$

e.g. for  $m = 3$

$$c_1 = (k_{11}p_1 + k_{12}p_2 + k_{13}p_3) \bmod 26$$

$$c_2 = (k_{21}p_1 + k_{22}p_2 + k_{23}p_3) \bmod 26$$

$$c_3 = (k_{31}p_1 + k_{32}p_2 + k_{33}p_3) \bmod 26$$

# Polyalphabetic Ciphers

- Basically multiple Caesar ciphers
- key is multiple letters long
  - $K = k_1 k_2 \dots k_d$
  - $i$ th letter specifies  $i$ th alphabet to use
  - mathematically give each letter a number

|   |   |    |    |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|----|----|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| a | b | c  | d  | e | f | g | h | i | j | k  | l  | m  | n  | o  | p  | q  | r  | s  | t  | u  | v  | w  | x  |
|   | y | z  |    |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 0 | 1 | 2  | 3  | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 |
|   |   | 24 | 25 |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

- Plaintext THISPROCESSCANALSOBEEEXPRESSED  
Keyword CIPHERCIPHERCIPHERCIPHERCIPHE  
Ciphertext VPXZTIQKTZWTCVPSWFDMTETIG AHLH

# Autokey Cipher

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- An **autokey cipher**, or **self-synchronizing stream cipher**, is a cipher which incorporates the message into the key.
- In order to encrypt a letter, one locates the line starting with the letter to be encrypted, and the column beginning with the next letter of the key. The letter where the line and column cross is the ciphertext letter.

For example, if the plaintext is MEET ME FOR LUNCH and the key is RESTAURANT, the first letter of the ciphertext is D, because the M and R intersect at D. The second ciphertext character is I, where the E from MEET and the E from RESTAURANT intersect in the table.

more secure way to generate the key is to append the plaintext to it. This method makes frequency counting more difficult, as the key does not follow a simple repeating pattern:

|    |    |    |    |    |    |    |    |    |   |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|---|----|----|----|----|----|----|----|
| a  | b  | c  | d  | e  | f  | g  | h  | i  | j | k  | l  | m  | n  | o  | p  | q  |
| r  | s  | t  | u  | v  | w  | x  | y  | z  |   |    |    |    |    |    |    |    |
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |   |    |    |    |    |    |    |    |

- Plaintext: MEETMEFORLUNCH
- Key: RESTAURANT**MEET**
- Ciphertext: DIWMMYWOEEGRGA

# One-Time Pad

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- In cryptography, the **one-time pad (OTP)** is an encryption algorithm in which the plaintext is combined with a secret random key or *pad*, which is used only once.
- The key is truly random, as large as the plaintext, never reused in whole or part, and kept secret.
- You have a message  $M$ , and you compute the ciphertext  $C$  by XORing the message with the OTP:
- $C = M \text{ xor OTP}$

# Transposition Ciphers

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- now consider classical **transposition** or **permutation** ciphers
- these hide the message by rearranging the letter order
- without altering the actual letters used

# Transposition Ciphers

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- A cipher, or a component of a more complicated cipher, that involves the symbol at each place in the plaintext being moved to a newly decided (and often different) place, i.e. within each block of plaintext the positional indices of the symbol places are changed according to some plan.

# Product Ciphers

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- ciphers using substitutions or transpositions are not secure because of language characteristics
- hence consider using several ciphers in succession to make harder, but:
  - two substitutions make a more complex substitution
  - two transpositions make more complex transposition
  - but a substitution followed by a transposition makes a new much harder cipher
- this is bridge from classical to modern ciphers

# Network Security

## Chapter 3 – “Block Ciphers and the Data Encryption Standard”



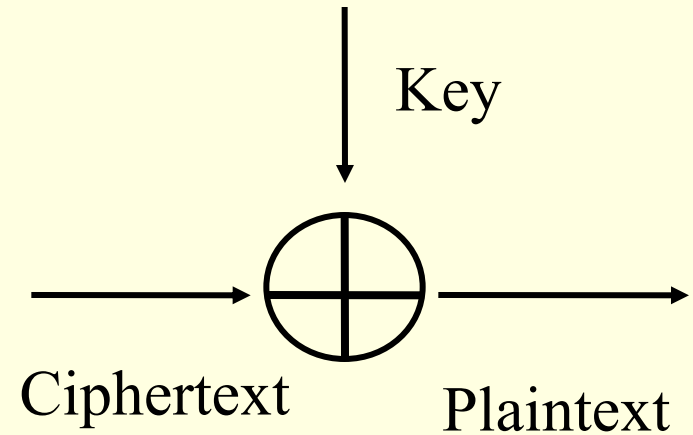
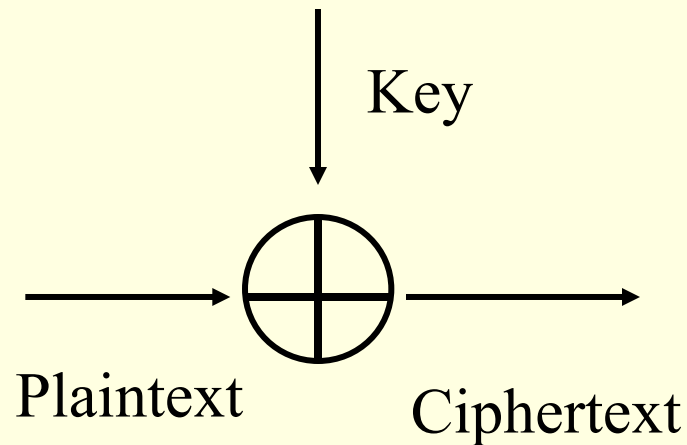
# Block Ciphers and the Data Encryption Standard

# Block vs Stream Ciphers

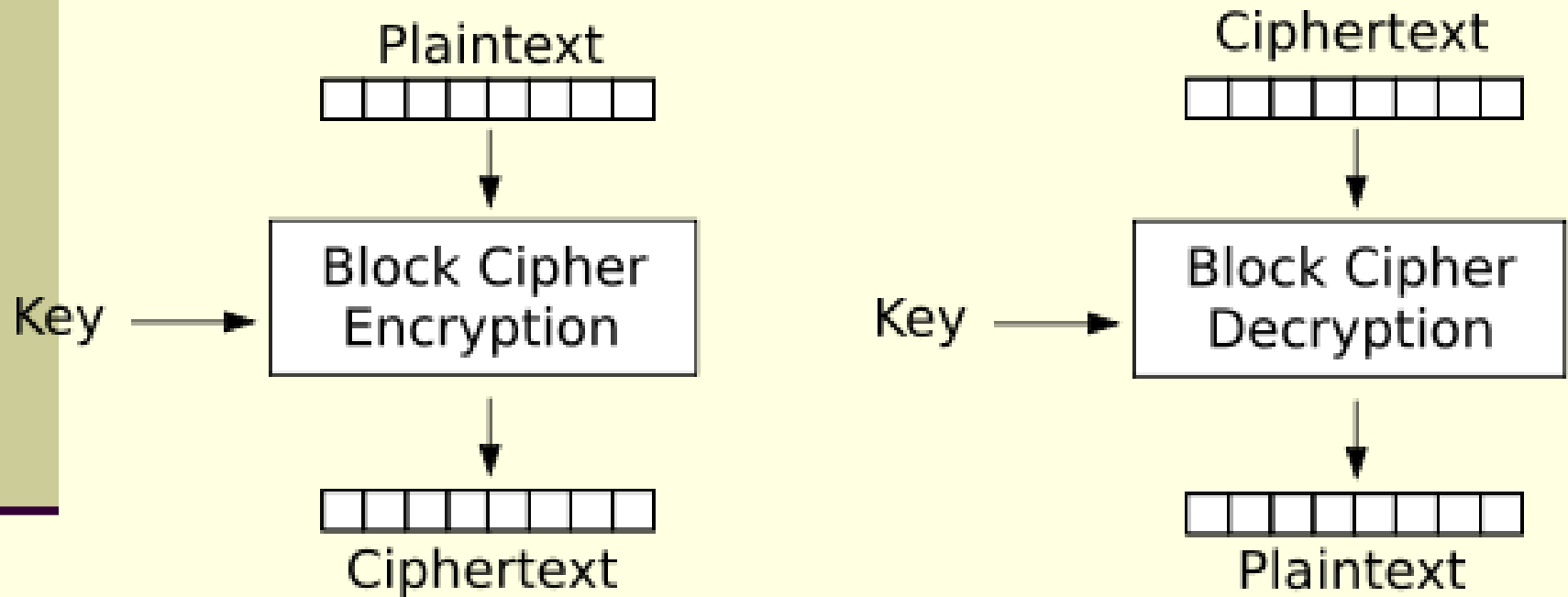
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- block ciphers process messages in blocks, each of which is then en/decrypted
- like a substitution on very big characters
  - 64-bits or more
- stream ciphers process messages a bit or byte at a time when en/decrypting
- many current ciphers are block ciphers
- broader range of applications

# Simplest Stream Cipher



# Block Cipher



# Block Cipher Principles

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- most symmetric block ciphers are based on a **Feistel Cipher Structure**
- needed since must be able to **decrypt** ciphertext to recover messages efficiently
- block ciphers look like an extremely large substitution
- would need table of  $2^{64}$  entries for a 64-bit block
- instead create from smaller building blocks
- using idea of a product cipher

# Claude Shannon and Substitution-Permutation Ciphers

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- Claude Shannon introduced idea of substitution-permutation (S-P) networks in 1949 paper
- form basis of modern block ciphers
- S-P nets are based on the two primitive cryptographic operations seen before:
  - *substitution* (S-box)
  - *permutation* (P-box)
- provide *confusion* & *diffusion* of message & key

# Confusion and Diffusion

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- cipher needs to completely obscure statistical properties of original message
- a one-time pad does this
- more practically Shannon suggested combining S & P elements to obtain:
- **diffusion** – dissipates statistical structure of plaintext over bulk of ciphertext
- **confusion** – makes relationship between ciphertext and key as complex as possible

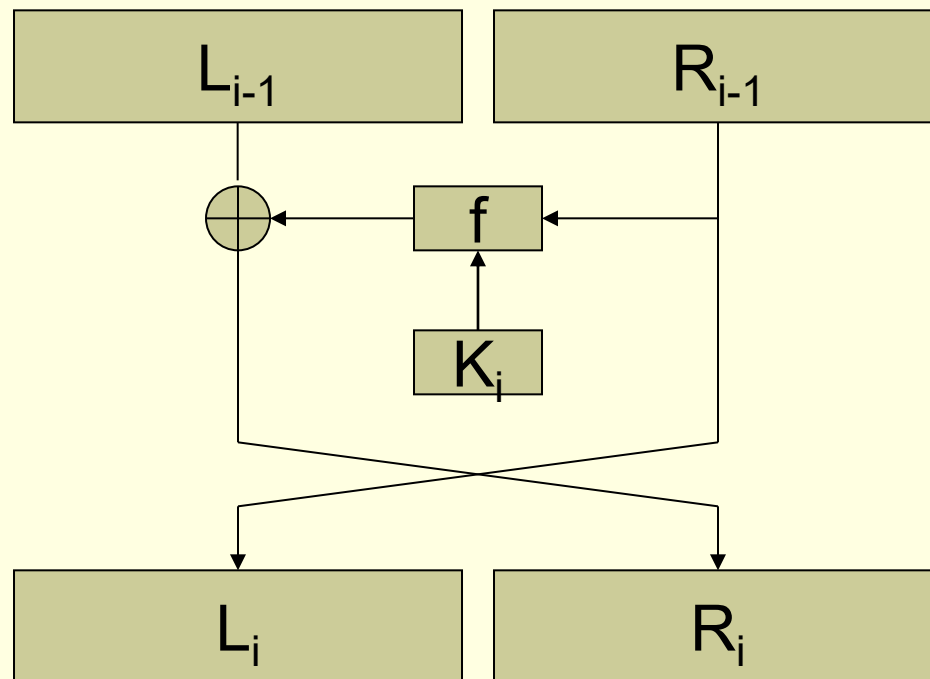
# Feistel Cipher Structure

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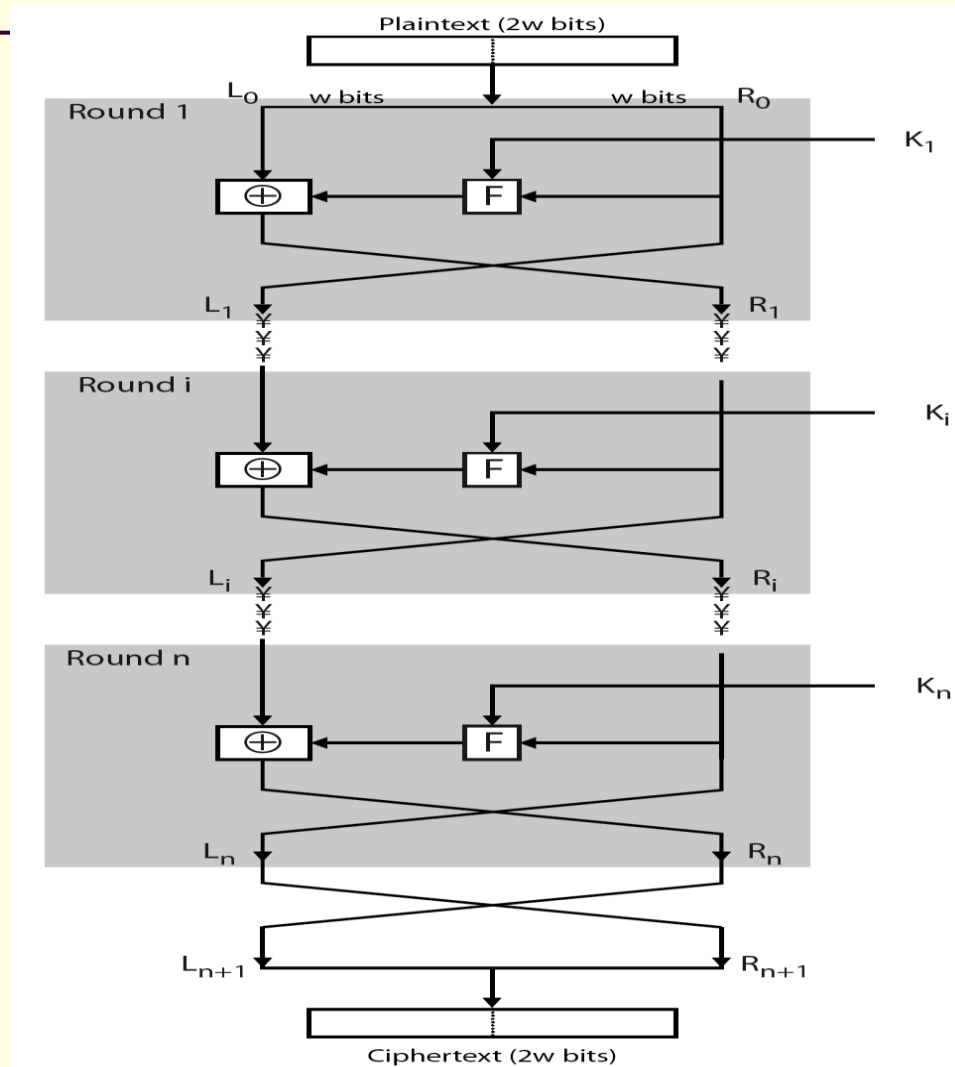
- Horst Feistel devised the **feistel cipher**
  - based on concept of invertible product cipher
- partitions input block into two halves
  - process through multiple rounds which
    - perform a substitution on left data half
    - based on round function of right half & subkey
    - then have permutation swapping halves
- implements Shannon's S-P net concept

# Feistel Network

- iterated cipher mapping  $(L_0, R_0)$  to  $(R_r, L_r)$  through  $r$ -round process,  $(L_{i-1}, R_{i-1}) \rightarrow_{K_i} (L_i, R_i)$  as follows
  - $L_i = R_{i-1}, R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$ ,  $K_i$  is derived from  $K$



# Feistel Cipher Structure

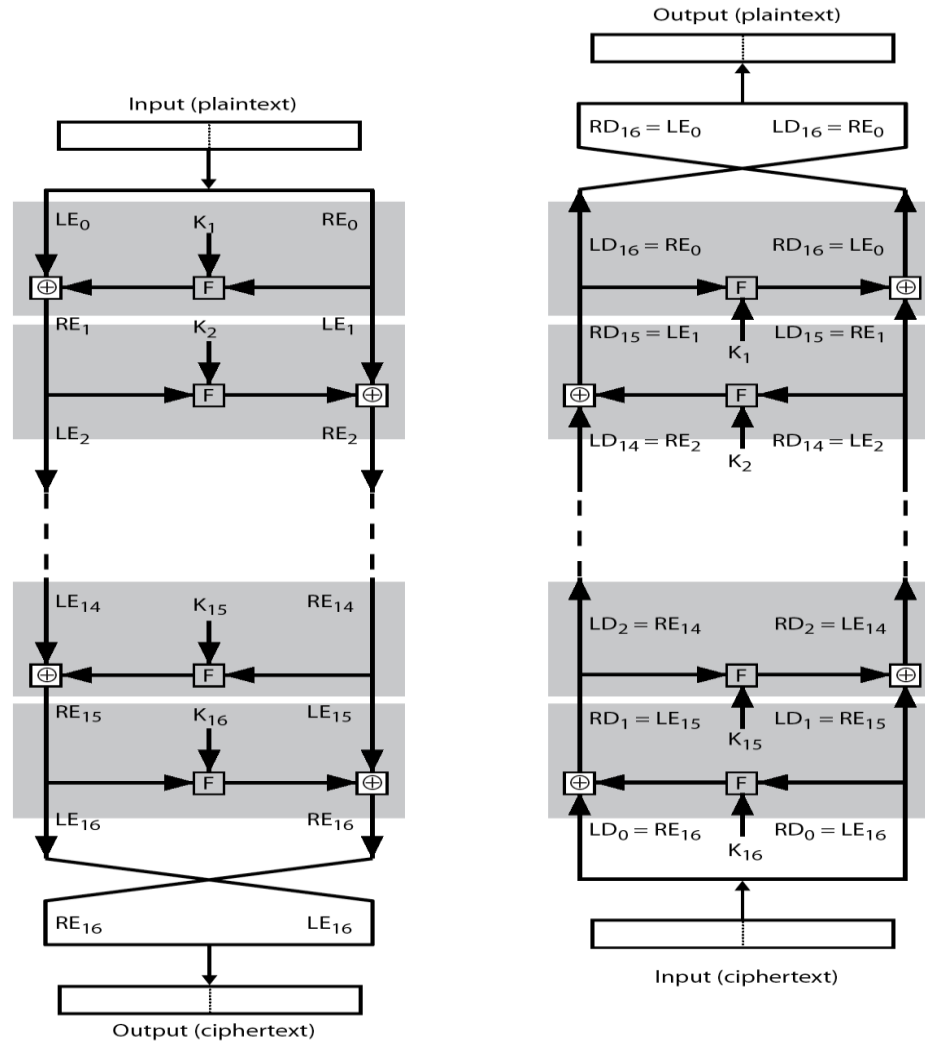


# Feistel Cipher Structure

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- **Block size:** larger block sizes mean greater security
- **Key Size:** larger key size means greater security
- **Number of rounds:** multiple rounds offer increasing security
- **Subkey generation algorithm:** greater complexity will lead to greater difficulty of cryptanalysis.
- **Fast software encryption/decryption:** the speed of execution of the algorithm becomes a concern

# Feistel Cipher Decryption

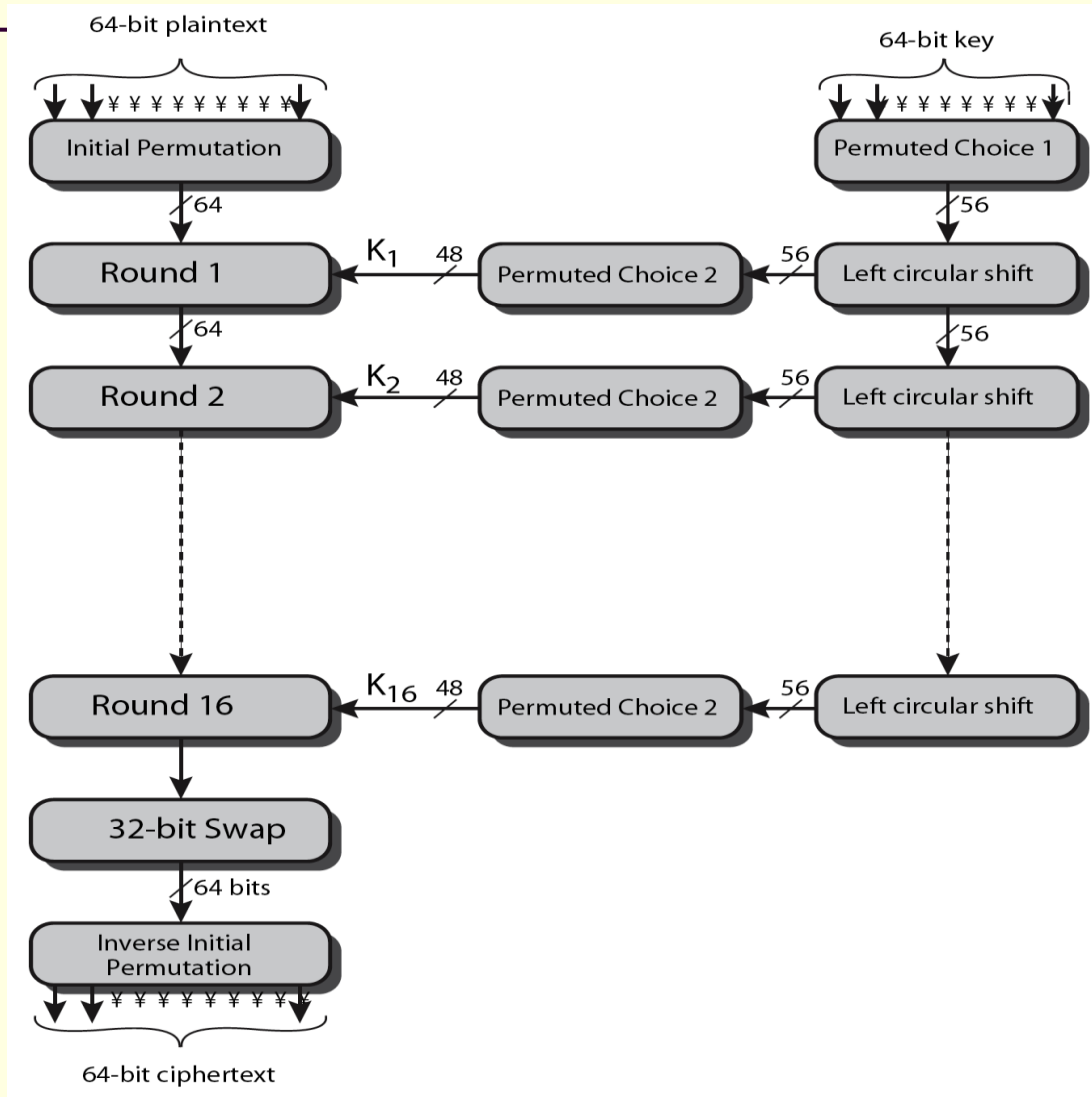


# Data Encryption Standard (DES)

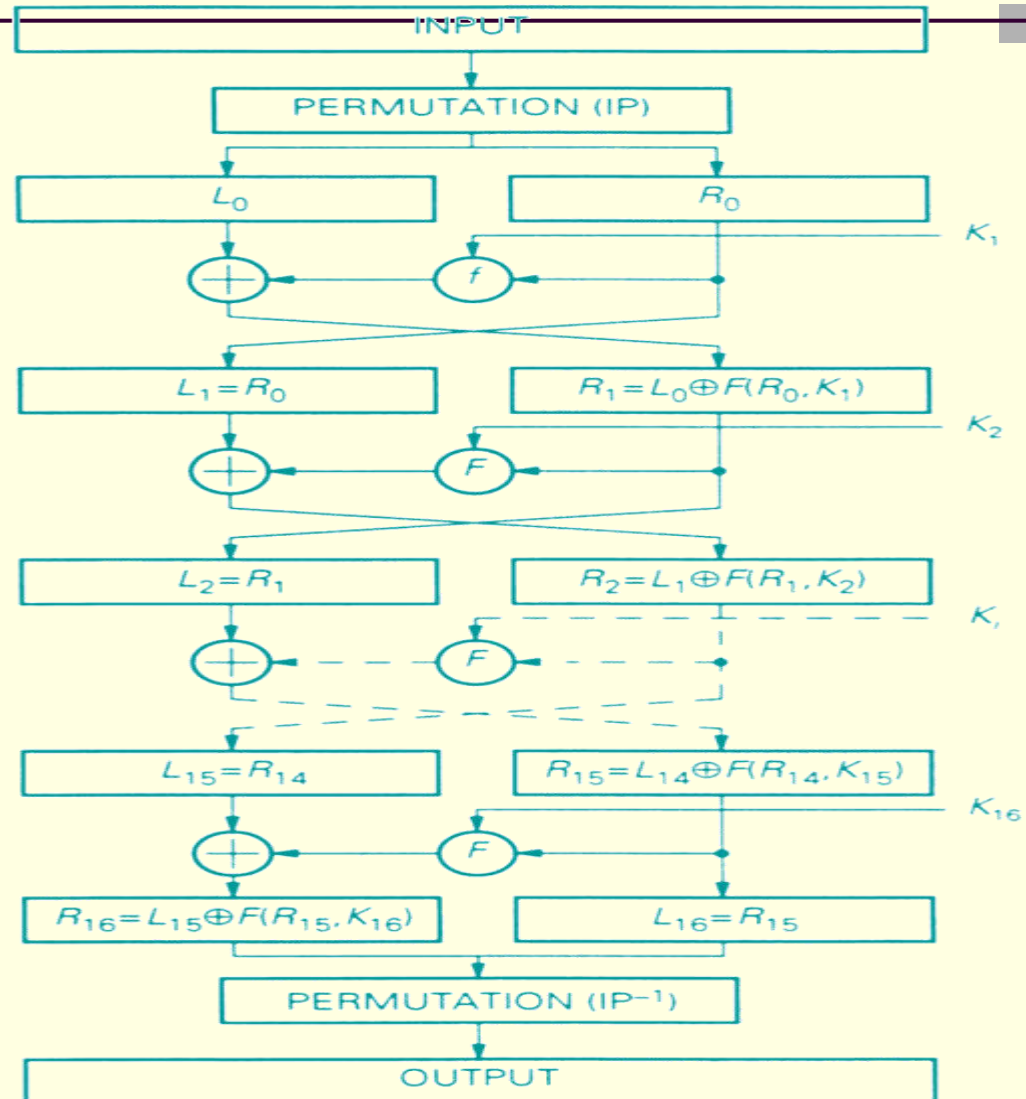
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- most widely used block cipher in world
- adopted in 1977 by NBS (now NIST)
  - as FIPS PUB 46
- encrypts 64-bit data using 56-bit key
- has widespread use
- has been considerable controversy over its security

# DES Encryption Overview



# DES



# DES Round Structure

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- uses two 32-bit L & R halves
- as for any Feistel cipher can describe as:

$$L_i = R_{i-1}$$

$$R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

- F takes 32-bit R half and 48-bit subkey:
  - expands R to 48-bits using perm E
  - adds to subkey using XOR
  - passes through 8 S-boxes to get 32-bit result
  - finally permutes using 32-bit perm P

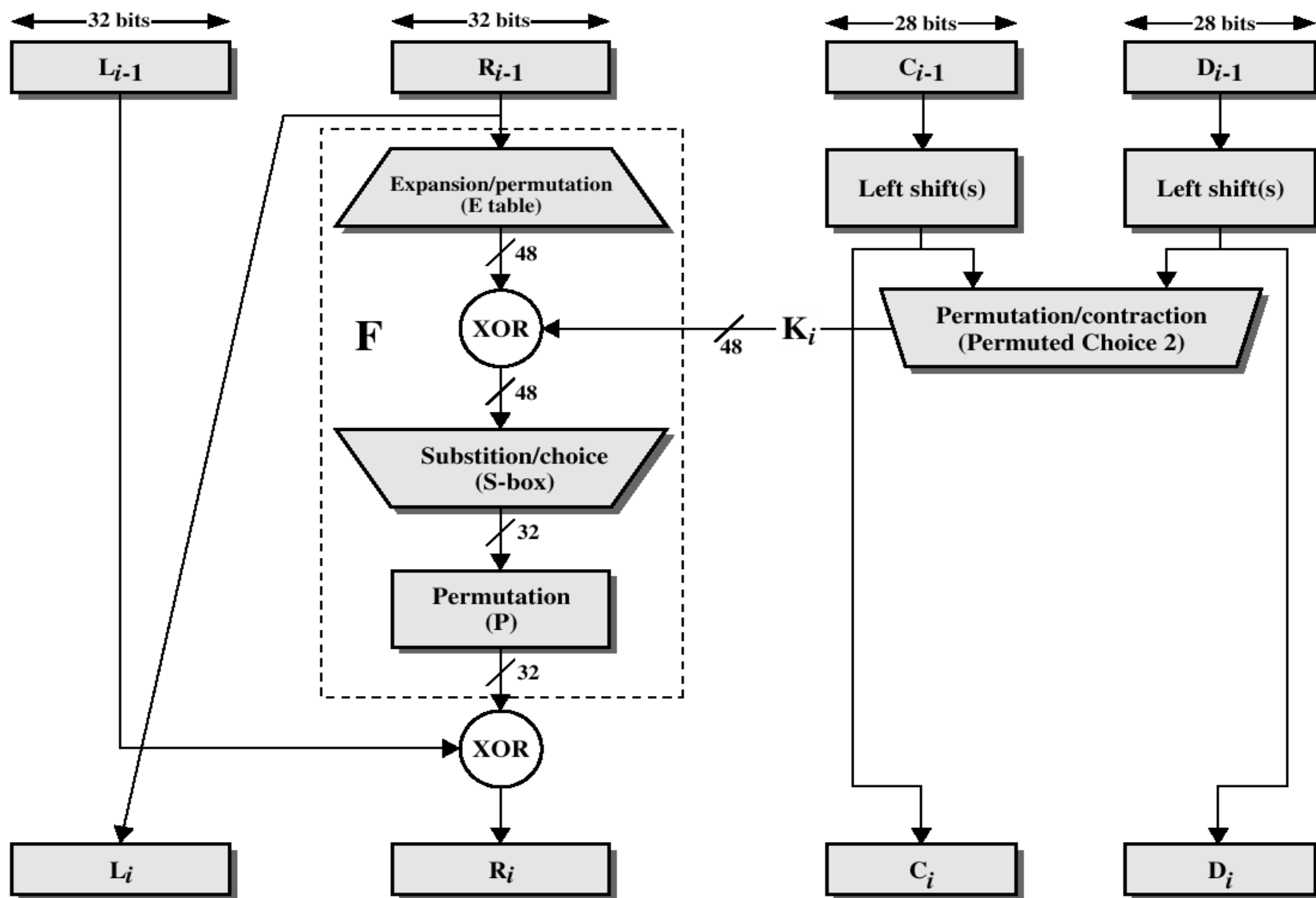


Figure 2.4 Single Round of DES Algorithm

# Initial Permutation IP

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- first step of the data computation
- IP reorders the input data bits
- even bits to LH half, odd bits to RH half
- The values in each matrix identify where each bit of the input message is mapped to in the output message. For example, The matrix for ***IP*** shows that the 58th bit from the input gets mapped to the first bit of the output; the 50th of the input maps to the second of the output, and so on.

| IP |    |    |    |    |    |    |   | IP <sup>-1</sup> |   |    |    |    |    |    |    |
|----|----|----|----|----|----|----|---|------------------|---|----|----|----|----|----|----|
| 58 | 50 | 42 | 34 | 26 | 18 | 10 | 2 | 40               | 8 | 48 | 16 | 56 | 24 | 64 | 32 |
| 60 | 52 | 44 | 36 | 28 | 20 | 12 | 4 | 39               | 7 | 47 | 15 | 55 | 23 | 63 | 31 |
| 62 | 54 | 46 | 38 | 30 | 22 | 14 | 6 | 38               | 6 | 46 | 14 | 54 | 22 | 62 | 30 |
| 64 | 56 | 48 | 40 | 32 | 24 | 16 | 8 | 37               | 5 | 45 | 13 | 53 | 21 | 61 | 29 |
| 57 | 49 | 41 | 33 | 25 | 17 | 9  | 1 | 36               | 4 | 44 | 12 | 52 | 20 | 60 | 28 |
| 59 | 51 | 43 | 35 | 27 | 19 | 11 | 3 | 35               | 3 | 43 | 11 | 51 | 19 | 59 | 27 |
| 61 | 53 | 45 | 37 | 29 | 21 | 13 | 5 | 34               | 2 | 42 | 10 | 50 | 18 | 58 | 26 |
| 63 | 55 | 47 | 39 | 31 | 23 | 15 | 7 | 33               | 1 | 41 | 9  | 49 | 17 | 57 | 25 |

*Table 2. Initial and final permutation matrices for DES.*

# Expansion Table

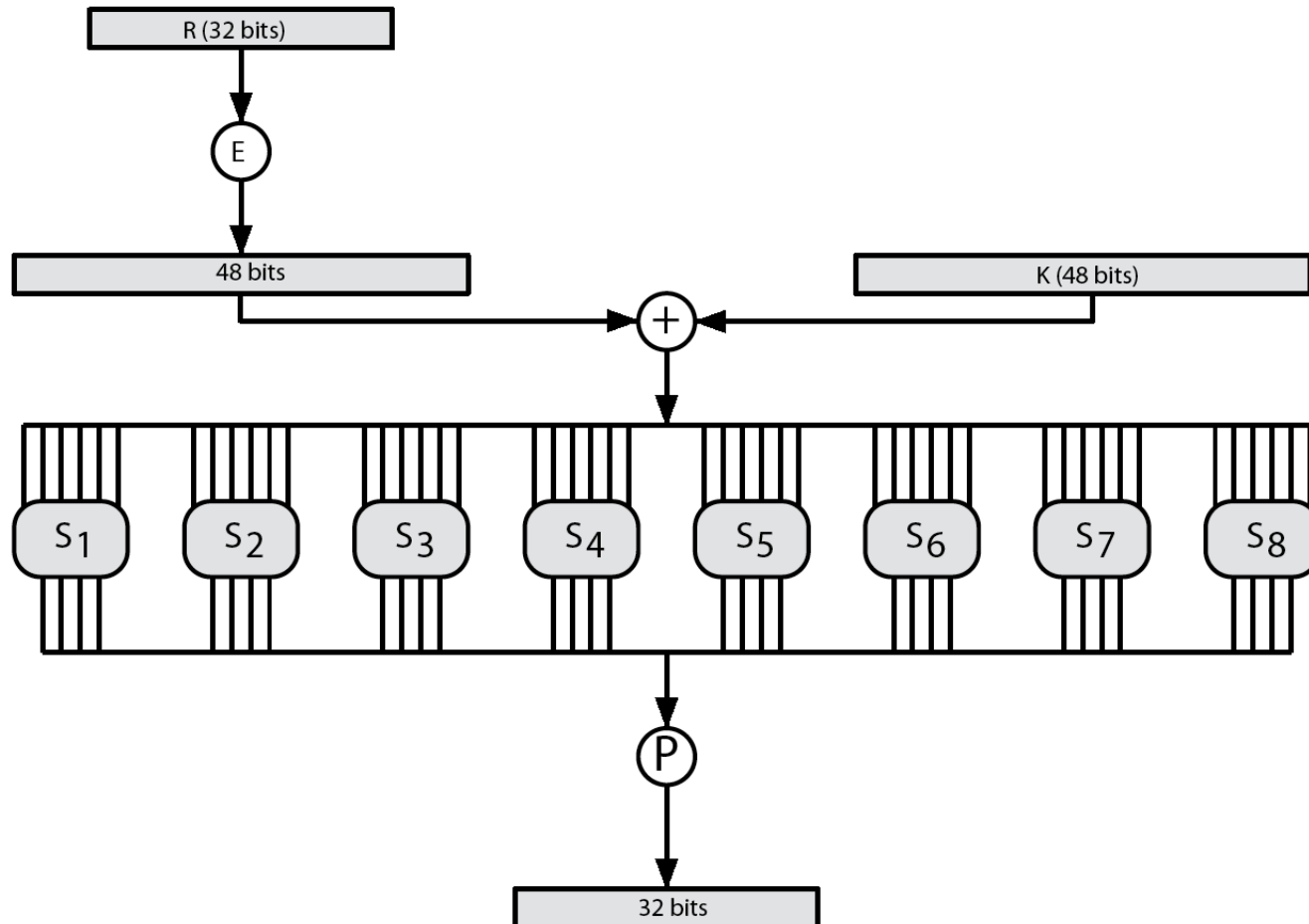
- Expands the 32 bit data to 48 bits
  - $\text{Result}(i) = \text{input}(\text{array}(i))$
  - The expansion table defines a permutation plus an expansion that involves duplication of 16 of the bits.

|    |    |    |    |    |    |
|----|----|----|----|----|----|
| 32 | 1  | 2  | 3  | 4  | 5  |
| 4  | 5  | 6  | 7  | 8  | 9  |
| 8  | 9  | 10 | 11 | 12 | 13 |
| 12 | 13 | 14 | 15 | 16 | 17 |
| 16 | 17 | 18 | 19 | 20 | 21 |
| 20 | 21 | 22 | 23 | 24 | 25 |
| 24 | 25 | 26 | 27 | 28 | 29 |
| 28 | 29 | 30 | 31 | 32 | 1  |

# S-Boxes

- The resulting 48 bits are XORed with key.
  - This 48 bit result passes through a substitution function comprising 8 S-boxes.
  - Which each map 6 input bits to 4 output bits.
- 
- Given 6-bits  $B = b_1b_2b_3b_4b_5b_6$ ,
    - Row  $r = b_1b_6$
    - Column  $c = b_2b_3b_4b_5$
    - $S(B) = S(r, c)$  written in binary of length 4

# DES Round Structure



# S-Box

- 6 bit input, 4 bit output
- $27 = \mathbf{011011} = (\mathbf{01})$   
(1101)
- $S_1$ -Box output for 27 = 5

|   | 0  | 1  | 2  | 3 | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 |
|---|----|----|----|---|----|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 14 | 4  | 13 | 1 | 2  | 15 | 11 | 8  | 3  | 10 | 6  | 12 | 5  | 9  | 0  | 7  |
| 1 | 0  | 15 | 7  | 4 | 14 | 2  | 13 | 1  | 10 | 6  | 12 | 11 | 9  | 5  | 3  | 8  |
| 2 | 4  | 1  | 14 | 8 | 13 | 6  | 2  | 11 | 15 | 12 | 9  | 7  | 3  | 10 | 5  | 0  |
| 3 | 15 | 12 | 8  | 2 | 4  | 9  | 1  | 7  | 5  | 11 | 3  | 14 | 10 | 0  | 6  | 13 |

# Permutation Table

- The permutation after each round

|    |    |    |    |
|----|----|----|----|
| 16 | 7  | 20 | 21 |
| 29 | 12 | 28 | 17 |
| 1  | 15 | 23 | 26 |
| 5  | 18 | 31 | 10 |
| 2  | 8  | 24 | 14 |
| 32 | 27 | 3  | 9  |
| 19 | 13 | 30 | 6  |
| 22 | 11 | 4  | 25 |

# DES Key Schedule

---

- The 64 –bit key input is first processed by permuted choice One.
- The resulting 56-bit key is then treated as two 28-bit quantities C and D.
- In each round forms subkeys
- rotating **each half** separately either 1 or 2 places depending on the **key rotation schedule K**
  - selecting 24-bits from each half & permuting them by PC2 for use in round function F
- note practical use issues in h/w vs s/w

# Permutation Tables

|    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|
| 57 | 49 | 41 | 33 | 25 | 17 | 9  |
| 1  | 58 | 50 | 42 | 34 | 26 | 18 |
| 10 | 2  | 59 | 51 | 43 | 35 | 27 |
| 19 | 11 | 3  | 60 | 52 | 44 | 36 |
| 63 | 55 | 47 | 39 | 31 | 23 | 15 |
| 7  | 62 | 54 | 47 | 38 | 30 | 22 |
| 14 | 6  | 61 | 53 | 45 | 37 | 29 |
| 21 | 13 | 5  | 28 | 20 | 12 | 4  |

Permutation table P1

■ After this permutation, the key is split into two halves, **C** and **D**. After each round, each half is independently shifted to the left by either one or two bits, depending on which round is executing. The shift is rotational, so that bits that get shifted off of one end get placed back on the other end.

|        |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |
|--------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|
| Round  | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| Shifts | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 2 | 1 | 2  | 2  | 2  | 2  | 2  | 2  | 1  |

*Table. Key Shifting Factors for each round of DES.*

Finally, the subkey function is used to convert the key into a 48 bit block, to be used in the actual encryption. Again, this is expressed in matrix form, as shown below.

### **Subkey Permutation**

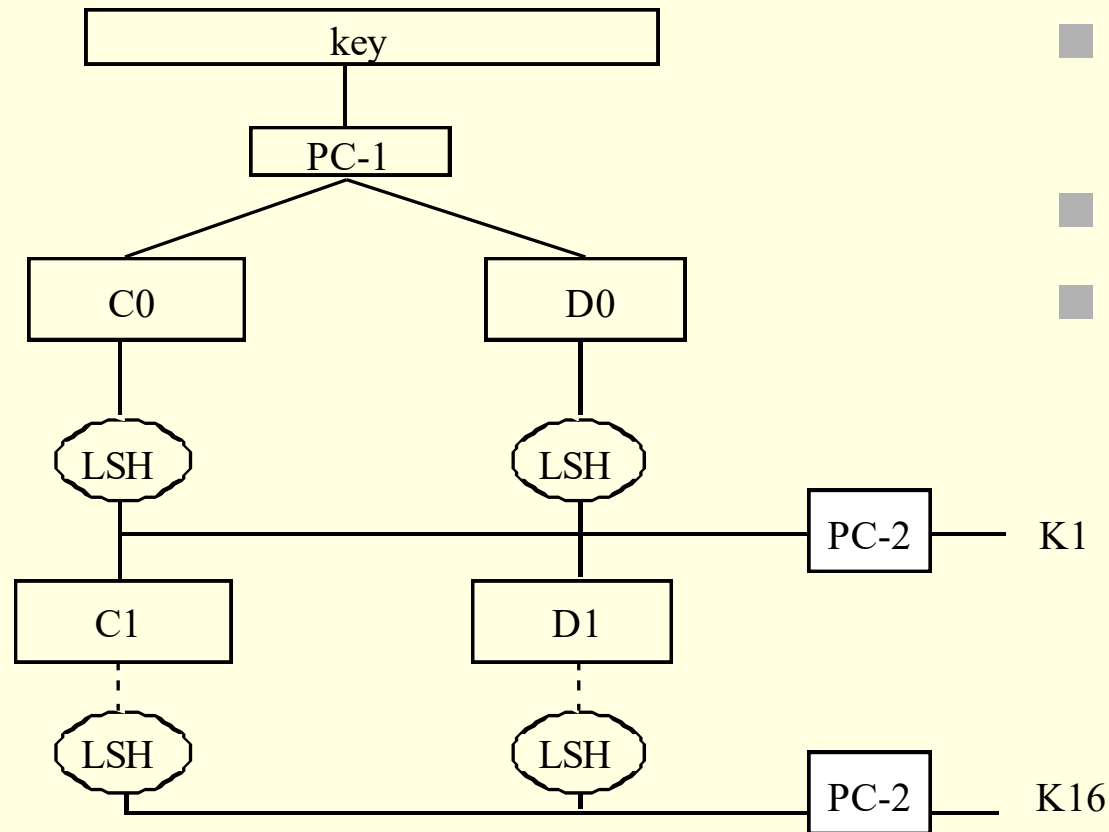
|    |    |    |    |    |    |
|----|----|----|----|----|----|
| 14 | 17 | 11 | 24 | 1  | 5  |
| 3  | 28 | 15 | 6  | 21 | 10 |
| 23 | 19 | 12 | 4  | 26 | 8  |
| 16 | 7  | 27 | 20 | 13 | 2  |
| 41 | 52 | 31 | 37 | 47 | 55 |
| 30 | 40 | 51 | 45 | 33 | 48 |
| 44 | 49 | 39 | 56 | 34 | 53 |
| 46 | 42 | 50 | 36 | 29 | 32 |

# DES Decryption

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- decrypt must unwind steps of data computation
- with Feistel design, do encryption steps again using subkeys in reverse order (SK16 ... SK1)
  - IP undoes final FP step of encryption
  - 1st round with SK16 undoes 16th encrypt round
  - ....
  - 16th round with SK1 undoes 1st encrypt round
  - then final FP undoes initial encryption IP
  - thus recovering original data value

# Generation of Round Keys



- Round keys are 48 bits each
- LSH == left shift
- PC-1, PC-2 == permutations

# Avalanche Effect

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- A desirable property of any encryption algorithm is that a small change in either the plaintext or the key should produce a significant change in the ciphertext.
- In particular, a change in one bit of the plaintext or one bit of the key should produce a change in many bits of the ciphertext.
- DES exhibits strong avalanche

# Strength of DES – Key Size

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- 56-bit keys have  $2^{56} = 7.2 \times 10^{16}$  values
- brute force search looks hard
- recent advances have shown is possible
  - in 1997 on Internet in a few months
  - in 1998 on dedicated h/w Electronic Frontier Foundation (EFF) in a few days
  - in 1999 above combined in 22hrs!
- still must be able to recognize plaintext
- must now consider alternatives to DES